

- 5 Traditional techniques for genetically modifying organisms use three enzymes:
- restriction endonuclease
 - reverse transcriptase
 - DNA ligase.

For example, these enzymes have been used to produce genetically modified (transgenic) pigs containing the *GFP* gene coding for green fluorescent protein, originally sourced from jellyfish.

- (a) Outline how these three enzymes could be used in genetically engineering a transgenic pig containing the *GFP* gene.

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..... [3]

A new technique that aims to cause a **deletion** in a gene uses an enzyme called Cas9 nuclease. It is injected into zygotes along with an RNA sequence (the guide RNA) that is complementary to a target gene. The Cas9 nuclease causes a deletion in the target gene in the zygotes, preventing the expression of that gene.

The toxicity and efficiency of the new technique was tested on four groups of pig zygotes. These pig zygotes were produced by IVF using:

- ova from a female non-transgenic pig.
- sperm from a male transgenic pig whose somatic (body) cells contained one copy of the *GFP* gene per cell.

The pig zygotes in three groups were injected with different concentrations of Cas9 nuclease and guide RNA **targeted at the *GFP* gene**.

The fourth group of pig zygotes (control group) was **not** injected with Cas9 nuclease and guide RNA.

- (b) Explain why the *GFP* gene was chosen for testing the new technique.

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Some of the zygotes in each group survived and after six days each had developed into a group of cells called a blastocyst.

The blastocysts were counted using a light microscope. A filter was then added to the microscope, so that only blastocysts expressing the green fluorescent protein showed up. These were counted and the results are summarised in Table 5.1.

Table 5.1

concentration of Cas9 nuclease and guide RNA / ng mm⁻³	number of blastocysts seen under white light	number of blastocysts seen under filter
0 (control)	68	46
10	40	0
20	24	0
50	15	0

- (c) (i) Calculate the percentage of zygotes in the control group that were transgenic.

Show your working.

..... % [1]

- (ii) Explain whether the percentage you calculated for (i) is higher or lower than expected.

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 [1]

- (iii) Name a statistical test that would allow you to test the significance of the difference between the percentage you calculated in (i) and the expected percentage.

..... [1]

- (iv) State the best concentration of Cas9 nuclease and guide RNA to use to cause a deletion in the *GFP* gene **and** give reasons for your choice.

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 [3]

(d) Fig. 5.1 shows the results from a second trial of the new technique, analysed by electrophoresis.

- Lanes **1–4** show DNA from four pigs born after Cas9 nuclease was used to cause a deletion in a target gene coding for a cell surface protein.
- Lane **5** shows DNA from their surrogate mother.
- Lane **6** shows DNA from another normal pig for comparison.

The size of the DNA fragments is given in kilobase pairs (kbp) as shown in Fig. 5.1. 1 kbp is 1000 base pairs of DNA.

The target gene measures 6kbp and codes for a cell surface protein that is essential for the disease virus PRRSV to infect cells in the pig's body.

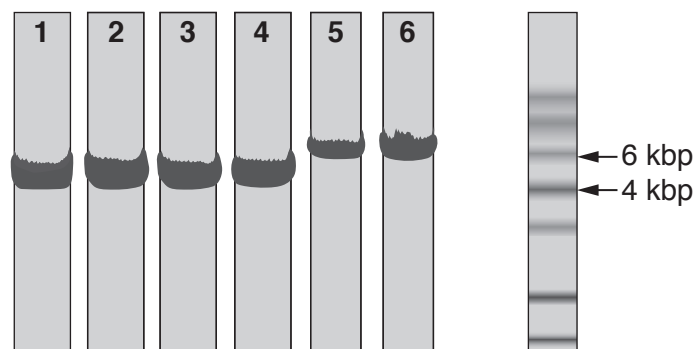


Fig. 5.1

Explain what Fig. 5.1 indicates about the success of the new technique in causing a deletion in a gene in pigs so that they show resistance to PRRSV.

[3]

[Total: 14]

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